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J.117

Pre-Junior Cycle Final Examination, 2025

Mathematics

Ordinary Level

Time: 2 hours

270 marks

CANDIDATE DETAILS

DAY and MONTH of BIRTH	__ __ / __ __ For example, 3rd February is entered as 03/02
NAME	
SCHOOL	
TEACHER	

School Stamp

For Examiner only

Question	Mark
1.	
2.	
3.	
4.	
5.	
6.	
7.	
8.	
9.	
10.	
11.	
12.	
Total	
%	

Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. You may ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1 **(Suggested maximum time: 5 minutes)**

(a) Find the value of each of the following.

(i) $574 + 419$

(ii) $4 \times 6{\cdot}3$

(iii) $28 \div (16 - 9)$

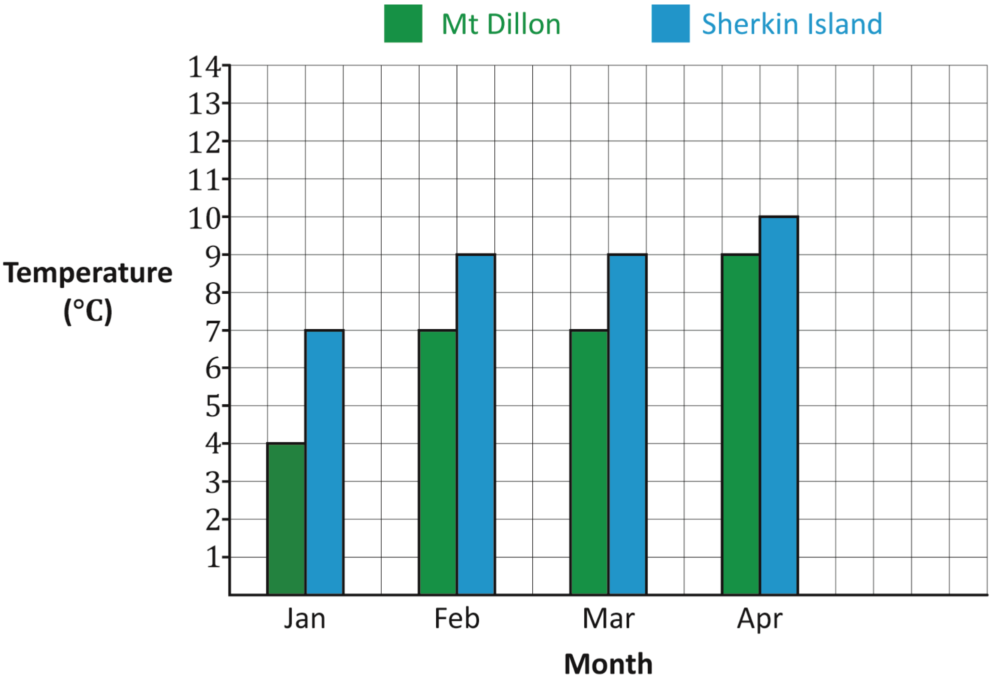
(b) **(i)** Work out how many **days** are in 17 weeks.

(ii) Work out how many **years** are in 192 months.

(iii) There was 53 Mondays in the year 2024.
What percentage of the total number of days in 2024 was made up of Mondays?
Give your answer correct to 1 decimal place.

Question 2 **(Suggested maximum time: 10 minutes)**

The average daily temperature (in °C) per month recorded at two weather stations, Sherkin Island and Mount Dillon, for the first four months of 2024 is shown in the diagram below.



(a) Use the diagram to answer the following questions.
In each case, tick (✓) the correct box only.

(i) The average daily temperature recorded in Sherkin Island during March was:

7 °C	9 °C	13 °C
☐	☐	☐

(ii) In Mount Dillon, the month with the lowest average daily temperature recorded is:

January	March	April
☐	☐	☐

(b) Work out the **difference** in the average daily temperature recorded at both weather stations during the month of January.

(c) The average daily temperature (in °C) recorded at the two weather stations for the month of May is shown in the table below. Complete the bar chart above to show this information.

Weather Station	**Sherkin Island**	**Mount Dillon**
Temperature (°C)	13	13

(d) Work out the **mean** daily temperature at Sherkin Island from January to May.

(e) A month is picked at random from the first 5 months of the year.
Tick the correct box to show the probability that the average daily temperature recorded at Mount Dillon weather station is above 8 °C.
Tick (✓) the correct box only.
Justify your answer.

0·2 ☐ 0·4 ☐ 0·8 ☐

Justification:

(f) The ratio of wet days to dry days on Sherkin Island in June 2024 was 2:1.
Work out the number of dry days recorded on Sherkin Island during June 2024.
(**Hint:** There are 30 days in June).

Question 3

(Suggested maximum time: 10 minutes)

(a) Nikita's gross pay is €$34\,000$ per year. His tax rate is 20% of his gross income.

(i) Work out Nikita's **gross tax** per year.

(ii) Nikita has a tax credit of €2750 per year. Work out Nikita's **net pay** per year.

(b) Nikita wins €$5\,000$ and decides to place it in a savings plan for 2 years.
The savings plan has an annual compound interest rate of 2%.

(i) Work out the interest he earned in the first year.

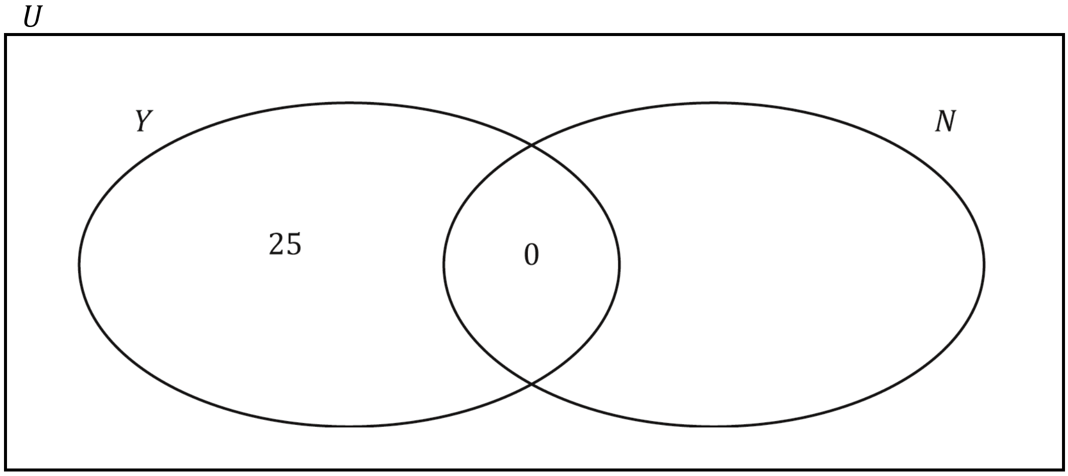
(ii) Work out the value of the savings plan at the end of 2 years.

$F = P(1+i)^t$

Question 4 **(Suggested maximum time: 5 minutes)**

Nabila asked all 48 students in her year group (U) if they were planning to do transition year (Y), or not planning to do transition year (N).

Some of the results are shown in her Venn diagram below.



(a) 6 students are undecided about whether to do Transition year and so did not answer the question. Fill this information into the Venn diagram above.

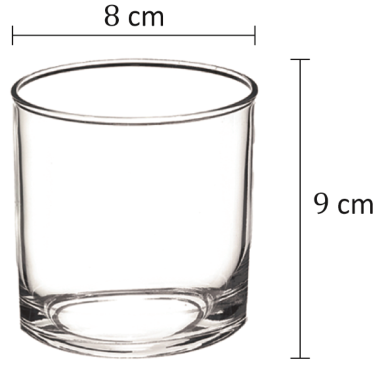
(b) Complete the Venn diagram above.

(c) A student is picked at random. What is the probability that they **have** decided on whether or not they are doing Transition year.

(d) Explain in your own words why $\#(Y \cap N) = 0$.

Question 5 **(Suggested maximum time: 15 minutes)**

(a) A glass is in the shape of a cylinder.
It has the dimensions shown in the diagram below.



(i) Write down the **radius** of the glass.

(ii) Work out the **volume** of the glass.
Give your answer correct to the nearest cm^3.

$V = \pi r^2 h$

(iii) Emma pours orange juice into the glass from a 500 cm^3 bottle until the glass is full.
Work out how much juice is left in the bottle.

(b) In 2024, Ireland introduced the deposit return scheme for plastic bottles, aluminium and steel cans. When you buy a drink in a container that features the Re-turn logo, you pay a deposit in addition to the price of the drink.

When you return your empty drink container, you get your deposit back.

Emma's bottle of juice cost €$2{\cdot}25$, **which included** the 15 cent deposit. Write the deposit paid as a fraction of the total cost of the bottle of coke. Give your answer in its simplest form.

Answer =

(c) The deposit cost depends on the size of the container:

Small containers from 150 ml to 500 ml have a deposit of 15c per item,
Large containers greater than 500 ml up to 3 litres have a deposit of 25c per item.

(i) Sean received a €3 refund when he returned a bag of small and large plastic bottles. Write down one possible combination of small and large bottles he may have returned.

Small bottles returned = # Large bottles returned =

(ii) Niamh received a €$2{\cdot}45$ refund when she returned 8 small plastic bottles and a number of large bottles. Work out the number of large bottles Niamh returned.

Question 6 **(Suggested maximum time: 10 minutes)**

Bryan is a Formula 1 fan.
He attended the British Grand Prix during his summer holidays.

(a) Bryan arrived at the Grand Prix at 10:50 a.m. and left at 6:25 p.m.
Tick the correct box to show how long Bryan spent at the Grand Prix.

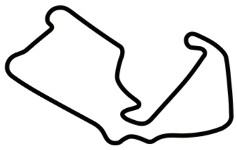
Tick (✓) **one** box only.
Justify your answer by suitable calculations.

7 hrs, 35 mins	6 hrs, 55 mins	8 hrs, 15 mins
☐	☐	☐

Calculations:

(b) Bryans ticket for the day cost £220.
The exchange rate at the time was €1 = £0·90.
Work out the cost of Bryan's ticket in euros.

(c) The Grand Prix race takes place over 52 laps of the Silverstone circuit.
The total length of the race is 306 km.
Work out the length of 1 lap of the track of the Silverstone circuit.
Give your answer in km, correct to 1 decimal place.



(d) Bryan's favourite driver, Lando Norris, completed the Grand Prix in 1 hour and 30 minutes.
Work out Lando Norris's **average speed** for the race in km/hr.

(e) The first 10 cars to finish the Grand Prix have the following car numbers:

44	1	4	81	55	27	18	14	23	22

Bryan said that **only one** of the car numbers is a prime number.
Is he correct?

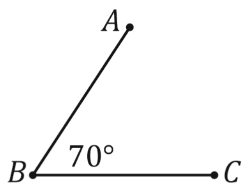
Tick (✓) **one** box only. Justify your answer.

Yes ☐ No ☐

Justification:

Question 7 **(Suggested maximum time: 10 minutes)**

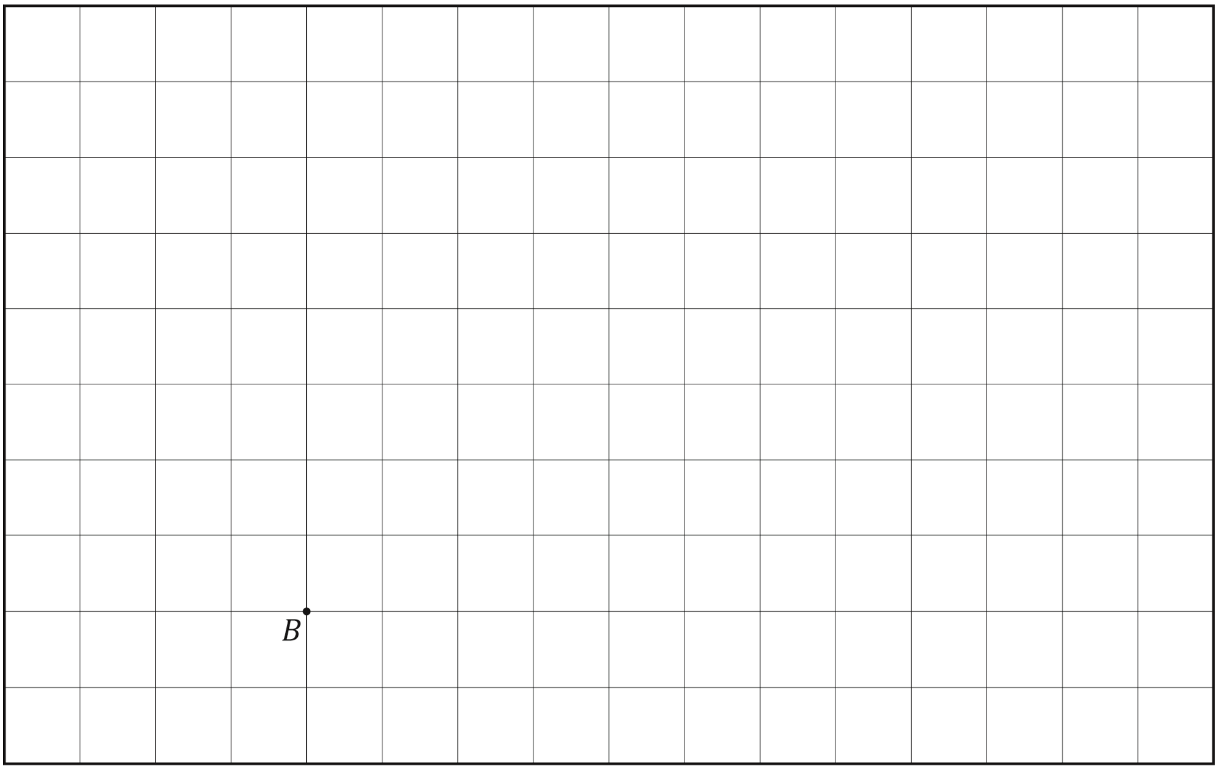
$|\angle ABC| = 70^\circ$ as shown (not to scale).



(a) Emilija bisects $\angle ABC$.
Work out the size of the two new angles she has created.

Hint: to bisect an angle means to divide it into exactly two equal parts.

(b) Draw the angle $\angle ABC$ in the space below.
The point B is marked for you.

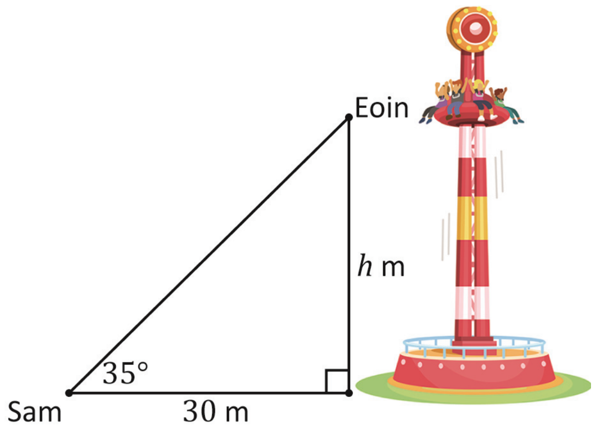


(c) Bisect this angle you have drawn above.
Show your construction lines clearly.

Question 8 **(Suggested maximum time: 5 minutes)**

Sam is at an amusement park.
He stands 30 m from the base of the vertical drop tower and sees his friend Eoin on the ride.

The angle of elevation from Sam to Eoin is $35°$.



(a) Use your calculator to find the value of $\tan 35°$.
Give your answer correct to 1 decimal place.

$\tan 35° =$ ☐

(b) Eoin is sitting at a height of h m from the base of the ride.
Use the right-angled triangle in the diagram to write $\tan 35°$ as a fraction, in terms of h.

$\tan 35° = \dfrac{\square}{\square}$

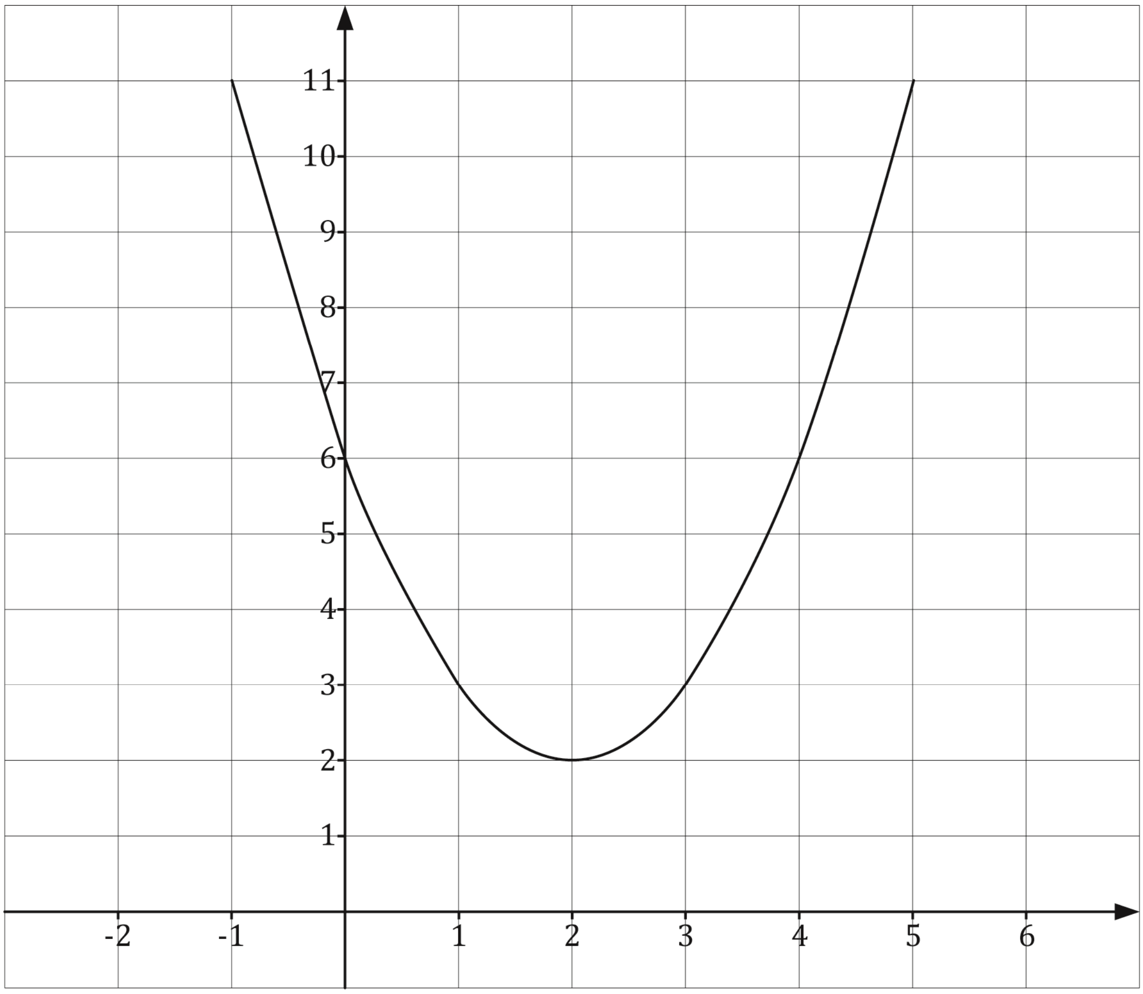
(c) Use trigonometry to work out the value of h.
Give your answer correct to the nearest metre.

Question 9 **(Suggested maximum time: 10 minutes)**

$f(x) = x^2 - 4x + 6$.

(a) Work out the value of $f(2)$.

(b) The graph of the function $y = f(x)$ for $-1 \leq x \leq 5$ and $x \in \mathbb{R}$ is shown on the co-ordinate diagram below.



(i) On the same axis, draw the graph of the linear function $h(x) = x + 2$ for $-1 \leq x \leq 5$ and $x \in \mathbb{R}$.

(ii) Use the graphs to write down the points of intersection of $f(x)$ and $h(x)$.

Answer = (,) and (,)

Question 10 **(Suggested maximum time: 5 minutes)**

(a) Tick the correct box to show whether the following pattern is linear or non-linear.

3 , 5 , 9 , 15 , 23 .

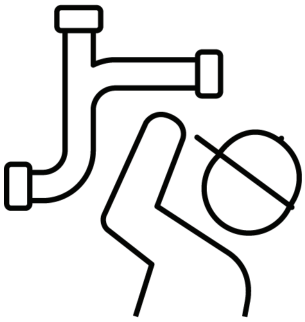
Tick (✓) **one** box only. Justify your answer.

Linear ☐ Non-linear ☐

Justification:

(b) A plumber charges a fixed call-out charge when called out to do a job. On top of the fixed call-out charge, there is an hourly charge rate for each hour worked after the first hour.

Time (hours)	Charge (€)
0	
1	
2	90
3	110



(i) Work out the hourly rate.

(ii) Work out the call-out charge

Question 11 **(Suggested maximum time: 10 minutes)**

(a) Evaluate $\dfrac{x-4}{2}-\dfrac{1}{3}$ when $x=5$.

(b) Solve the following equation in x:

$$8x+19=3x+34.$$

(c) Multiply out and simplify $(y-7)(y+4)$.

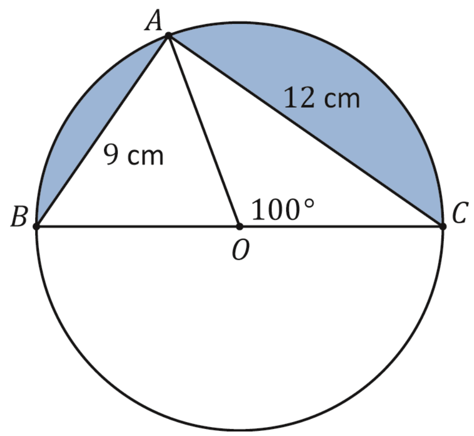
(d) Factorise $x^2+12x+35$.

$x^2+12x+35 = (x \qquad)(x \qquad)$

Question 12 **(Suggested maximum time: 15 minutes)**

The diagram below shows a triangle standing on the diameter $[BC]$ of a circle, centre O.

$|AB| = 9$ cm, $|AC| = 12$ cm and $|\angle AOC| = 100°$.



(a) Tick the correct box to show $|BC|$.
Tick (✓) **one** box only.
Use the Theorem of Pythagoras to justify your answer.

13 cm ☐ 15 cm ☐ 25 cm ☐

Justification:

(b) Tick the correct box to show $|OA|$, the length of the radius of the circle.
Tick (✓) **one** box only.

4·5 cm ☐ 6 cm ☐ 7·5 cm ☐

(c) Identify two **isosceles triangles** in the diagram above.

Isosceles triangles: Δ ______ and Δ ______

(d) Find the size of the angle $\angle AOB$ and the angle $\angle OBA$, without measuring.

$|\angle AOB| =$ ______ $|\angle OBA| =$ ______

(e) Work out the **area** of the **circle**, centre O in the diagram.
Give your answer in cm^2, correct to 1 decimal place.

$A = \pi r^2$

(f) Work out the **area** of the **triangle** ABC.

(g) Work out the **area** of the **shaded region** in the diagram.

Do not write on this page

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